(a) HiLoStem Encoder

$X \in R^{3\times H\times W}$



Input

Detail-Preserving (High-Resolution Branch)

3×3 Conv, s=1 → 7×7 DWConv, s=1 → SE (Gate)

y

Downsample
AvgPool 4×4, s=4

$\tilde{y}$

Patch Embedding (Low-Resolution Branch)

4×4 Conv, s=4

P

E

Tokenization
(Flatten to Tokens)

Token Sequence
(N × C)

Element-wise addition

HiLoStem components

(b) Swin Transformer Backbone

Swin Stage 1 (*H*/4 × *W*/4) → *F1*

Swin Stage 2 (*H*/8 × *W*/8) → *F2*

Swin Stage 3 (*H*/16 × *W*/16) → *F3*

Swin Stage 4 (*H*/32 × *W*/32) → *F4*

(c) Geometry-Aware Decoder

Proj 3×3 (×4)

Decoder Stage 1 (*H*/4 × *W*/4) Refinement

Decoder Stage 2 (*H*/8 × *W*/8) UP + WF

Decoder Stage 3 (*H*/16 × *W*/16) UP + WF

Decoder Stage 4 (*H*/32 × *W*/32) UP + WF

AxialLite (Cross-Region Mixer)

Final Decoder Feature
F *(H × W)*

BD Head

Boundary Map $B \in R^{1\times H\times W}$

Distance Field $D \in R^{1\times H\times W}$

Head

SegHead (1 × 1 Conv)

Logits $\hat{Y}$ *(C × H × W)*

$\hat{Y} \in R^{H\times W}$



Output

Skip Connection

Training-only

UP Upsampling

WF Weighted Fusion